Yet another epsilon meteor echo simulator

An interactive phenomenological model of overdense forward-scatter meteor echoes
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Abstract

Overdense meteor forward-scatter echoes can display curved and rapidly evolving Doppler signatures. This contribution describes *echoSim.py*, an interactive Python simulator developed as a compact experiment for exploring how an extended meteor trail, bistatic observing geometry and a parameterised atmospheric wind field can combine to produce such signatures. The implementation defines an initially skewed trail, constructs the transmitter–trail–receiver geometry for every trail element, identifies the minimum total path as the specular location, and uses the local bistatic bisector to project the evolving velocity into Doppler frequency. A multi-harmonic wind field deforms the trail, while a slope-dependent weighting controls the relative contribution of different trail elements. The signal is generated by continuous phase accumulation and coherent summation, followed by a prescribed event envelope and optional Gaussian noise. The simulator is intentionally phenomenological; its purpose is to provide a transparent environment for testing ideas about Doppler morphology rather than to provide a validated inversion model.

1. Introduction

Meteor forward-scatter echoes are governed by the ionised trail, bistatic geometry and motion of the scattering plasma. In overdense events, an extended trail can contribute at multiple Doppler frequencies, while the effective scattering region evolves. echoSim.py is designed as a controllable numerical experiment rather than a complete plasma-physics model.

2. Model

The initial trail is x■(z) = tan(α)z, with z spanning −1.2 to +1.2 km and default α = 12.0°. Tx and Rx are separated by 50 km, with a reference altitude of 90 km. For each trail element, the code computes r■, r■, the unit propagation vectors and their sum, which defines the local bistatic bisector. The specular element minimises r■+r■. Wavelength is λ=c/f.

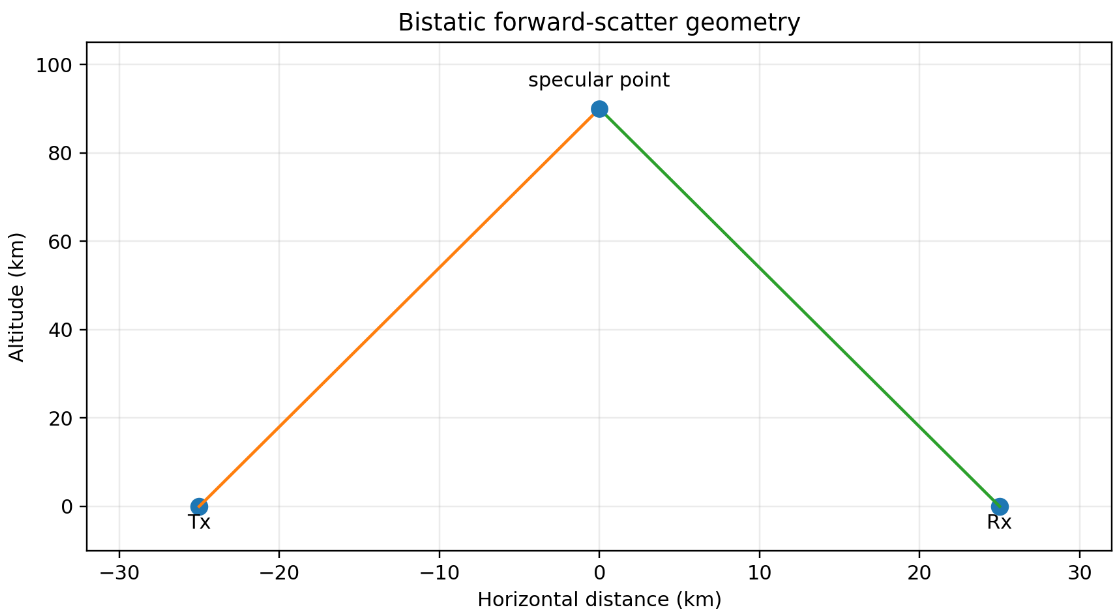

Figure 1. Simplified bistatic geometry.

The wind field is v(z)=V[R■sin(πz)+R■sin(2πz)+R■sin(3πz)], with default V=80 m/s, R■=1.00, R■=0.00, R■=0.10. Its derivative drives trail deformation. The field is phenomenological.

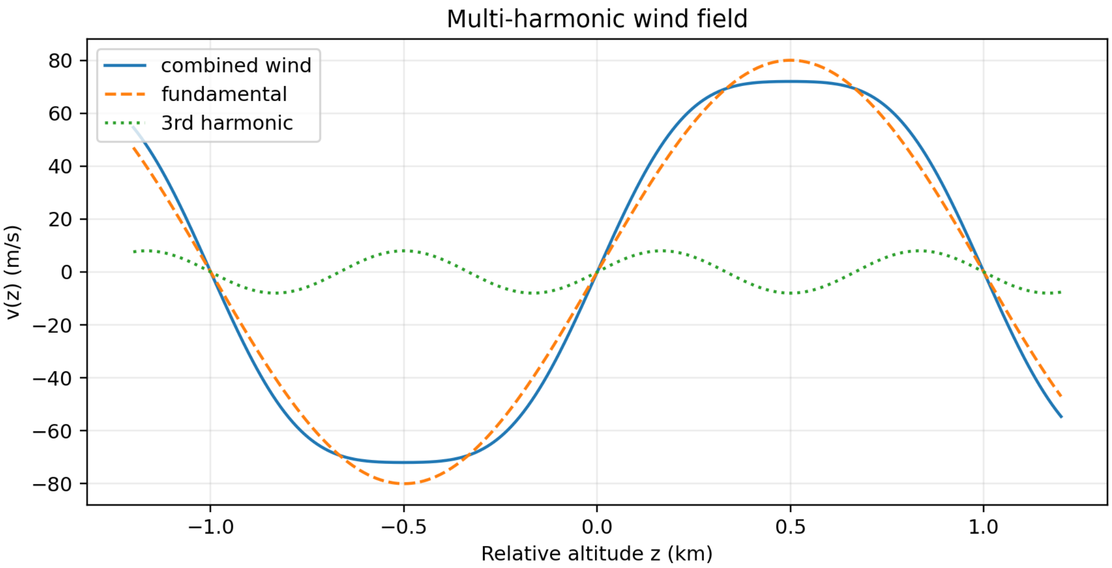

Figure 2. Example multi-harmonic wind field.

The ping-like velocity contribution decays as $\exp(-\tau/0.12)$, while the wind contribution grows as $1-\exp(-\tau/0.40)$. Deformation is proportional to τ times the wind weight and the implementation coefficient 0.0025.

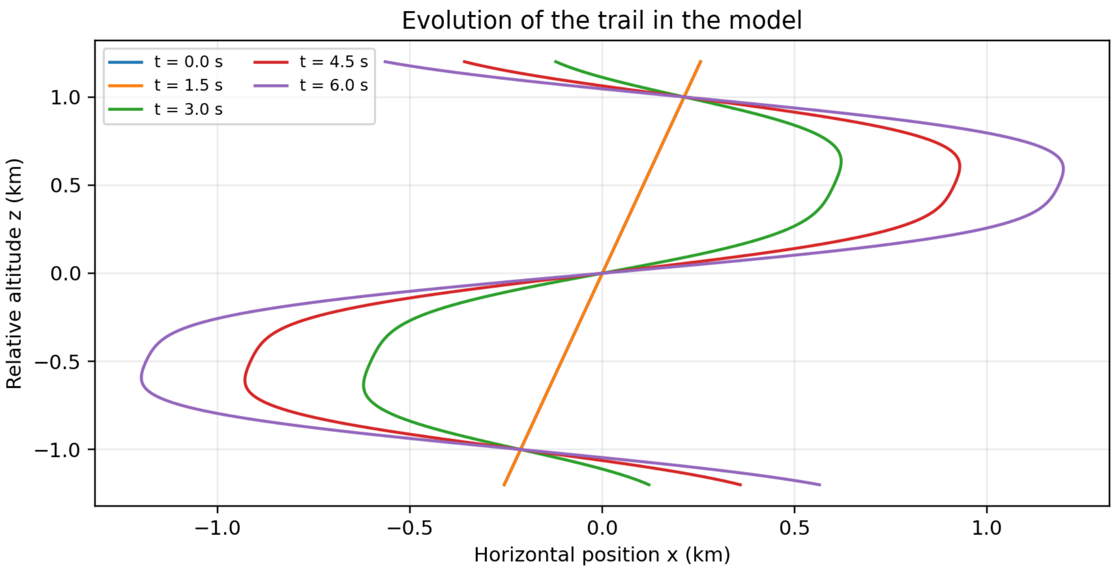

Figure 3. Illustrative trail evolution.

The local Doppler offset is $f_D(z,t)=v(z,t)\cdot b(z)/\lambda$. Trail elements are weighted by $W=\exp[-(\partial x/\partial z)^2/A(t)]$. Their instantaneous 1 kHz audio frequencies are integrated as phase and summed coherently. The echo starts at t=2 s; diffusion/decay begins at t=3 s. Optional Gaussian noise is then added.

3. Interactive implementation

The GUI controls Tx–Rx distance, specular altitude, RF frequency, trail skew, wind speed, three harmonic ratios, azimuth and elevation. It can add noise, play audio and export WAV and PNG files.

4. Example

The default configuration is d=50 km, h=90 km, f■=50.0 MHz, skew=12.0°, wind=80 m/s, R■=1.00, R■=0.00, R■=0.10, az=180°, el=45°. The example is illustrative, not a fit to a particular observation.

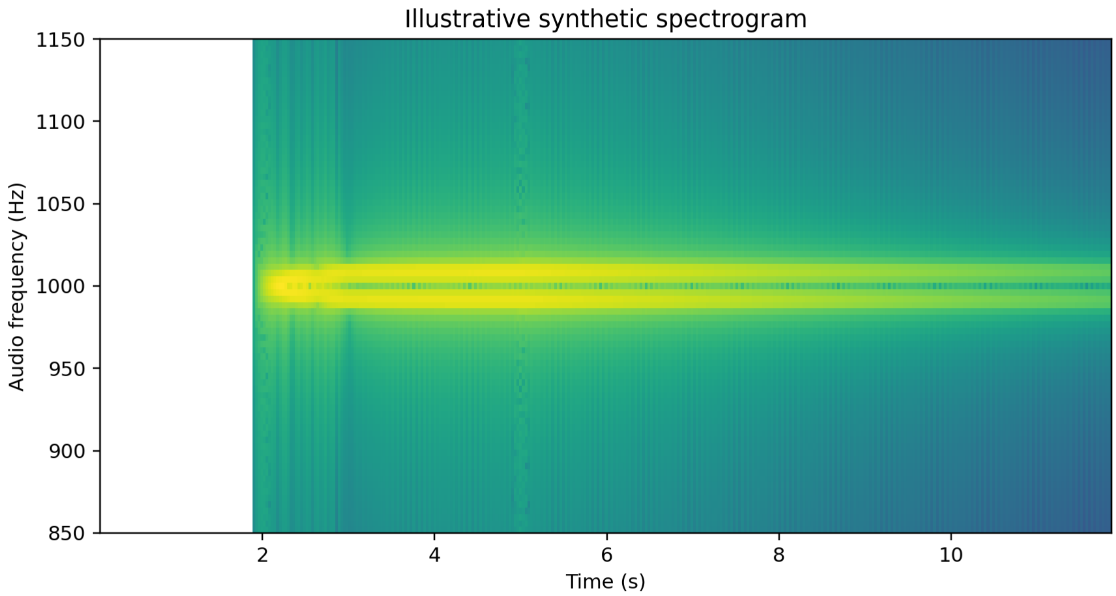

Figure 4. Illustrative synthetic spectrogram.

5. Discussion and limitations

The model is transparent but several components are phenomenological: the harmonic wind field, event envelope, deformation coefficient and specularity aperture. It should therefore not yet be treated as a physical inversion model. Its immediate purpose is controlled exploration of Doppler morphology.

6. Future work

Future work should include measured or modelled wind profiles, quantitative comparison with observed spectrograms, explicit trail density and diffusion, improved scattering efficiency and parameter-sensitivity studies.

7. Conclusions

echoSim.py combines explicit bistatic geometry, an initially skewed extended trail, multi-harmonic wind shear, trail deformation, slope-dependent specularity and coherent phase summation. It provides a compact test bench for investigating epsilon-like overdense meteor echo signatures; validation against observations is the essential next step.